```
architecture MUX221 of MUX2to1 is

begin
q<=I0 when S0='0' else
  I1 when S0='1' else
  'Z';
end MUX221;
```

The code below shows a 4:1 multiplexer:

```
--4:1 MUX-MUX4to1.vhd
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity MUX4to1 is
  Port (I0,I1,I2,I3,S0,S1 : in STD_
        LOGIC;
    q : out STD_LOGIC);
end MUX4to1;

architecture M421 of MUX4to1 is
signal sel:STD_LOGIC_VECTOR(1 downto 0);
begin
sel<=(S1,S0);
q<=I0 when sel="00" else I1 when
   sel="01" else
   I2 when sel="10" else I3 when
   sel="11" else
   'Z';
end M421;
```

The UniShiftReg.vhd code implements VHDL code for universal shift register.

Register file

Data path has a set of addressable registers whose control signals and input and outputs are interfaced to common input and output ports, respectively. Such a set is referred to as register file. Width of the register is set by generic 'width.' Address width is 'n,' resulting in $2^n = m$ registers. There is one input port and two output ports (Port A and Port B). Write address chooses the register to which input word is written at the falling edge of the clock. Write address is decoded by the address decoder. Port A receives the output of the register referred to by Port-A address. Port B receives the output of the register referred to by Port-B address. Two address decoders are used to read a word. Each register consists of multiple D-type flip-flops. D-type flip-flop is modified to include Enable signal that gates the clock input. The DFFEn.vhd code implements D-type flip-flop with Enable signal.

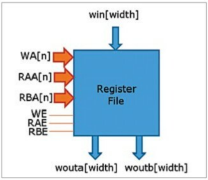

Fig. 9: Register file

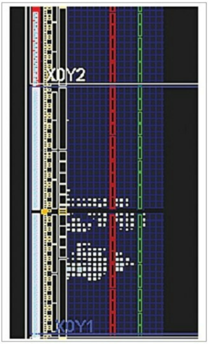

Fig. 10: Data implementation

EFY Note
The source codes of this project are included in this month's EFY DVD and are also available for free download at *source.efymag.com*

Address decoder has 'n'-bit-wide input and $2^n = m$ bit output. All the output bits are '0,' except the one referred to by the input, which is '1.' The AddrDec.vhd code implements VHDL code for the address decoder.

Multiple D-type flip-flops can be used to store a word. Such a circuit is referred to as the register. The Reg.vhd code implements VHDL code for a register. Block diagram of the register file is shown in Fig. 9. The RegFile.vhd code implements VHDL code for register file.

Data path integration

Now that you have all the components of a data path as per Fig. 2, write a structural-style VHDL code for data path as given below:

```
--Data Path -DataPath.vhd
library IEEE; use IEEE.STD_LOGIC_1164.
ALL;
library upcomp; use upcomp.SubComp.all;
entity DataPath is
..................................
..................................
end DP;
```

Fig. 10 shows the implementation of data path on FPGA. Fig. 11 shows the schematic of data path.

Data path testing

Data path requires a large number of I/Os, which Nexys 4 board


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